Data Mining

Evaluation Metrics

Topics

- Why are metrics important?
- Binary classifiers
  - Rank view, Thresholding
- Metrics
  - Confusion Matrix
  - Point metrics: Accuracy, Precision, Recall / Sensitivity, Specificity, F-score
  - Summary metrics: AU-ROC, AU-PRC, Log-loss.
- Choosing Metrics
- Class Imbalance
  - Failure scenarios for each metric
- Multi-class

Why are metrics important?

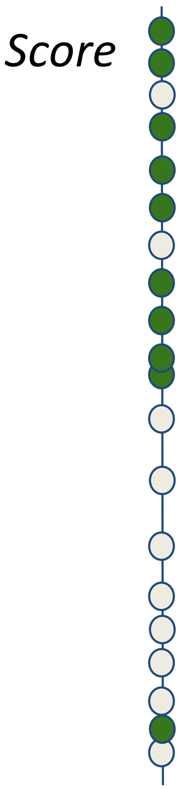
- Training objective (cost function) is only a proxy for real world objectives.
- Metrics help capture a business goal into a quantitative target
- Helps organize effort towards that target.
  - Generally in the form of improving that metric on the dev set.
- Useful to quantify the “gap” between:
  - Desired performance and baseline (estimate effort initially).
  - Desired performance and current performance.
  - Measure progress over time.
- Useful for lower level tasks and debugging (e.g. diagnosing bias vs variance).

Binary Classification

- x is input
- y is binary output (0/1)
- Model is $\hat{y} = h(x)$
- Two types of models
  - Models that output a categorical class directly (K-nearest neighbor, Decision tree)
  - Models that output a real valued score (SVM, Logistic Regression)
    - Score could be margin (SVM), probability (LR, NN)
    - Need to pick a threshold
    - We focus on this type (the other type can be interpreted as an instance)

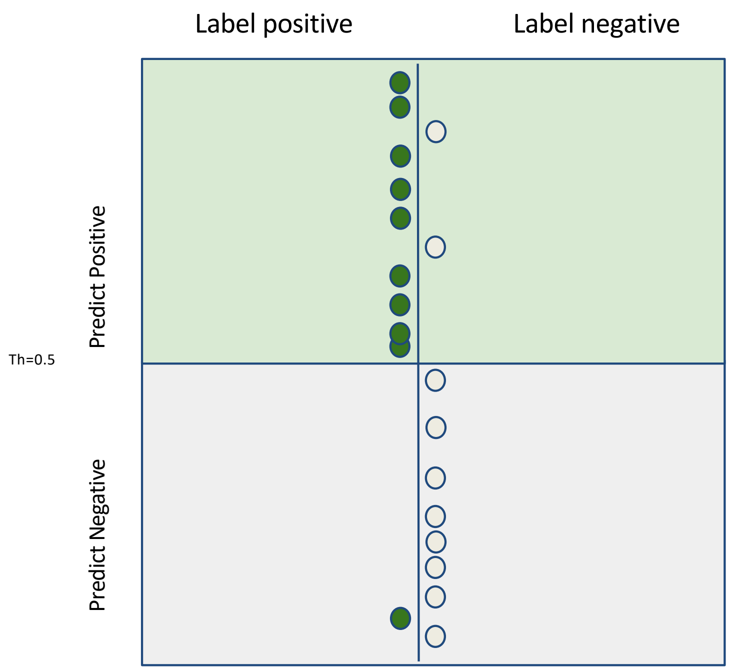
Score based models

Example of Score: Output of logistic regression



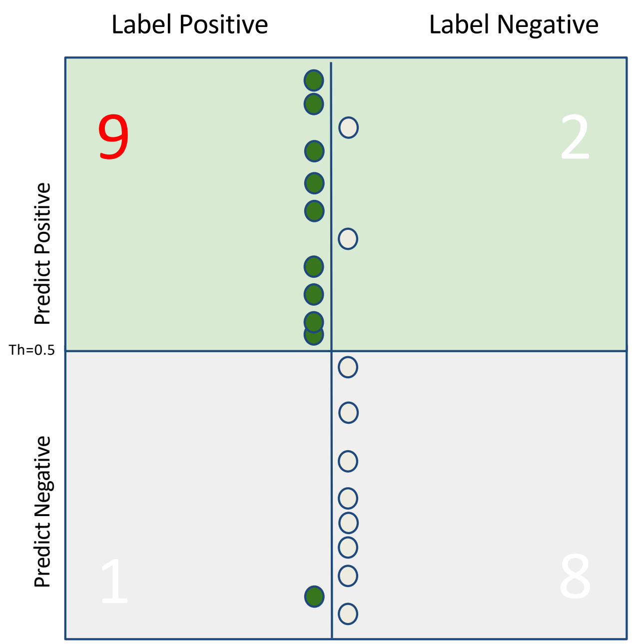
●	Positive example
○	Negative example

Threshold → Classifier → Point Metrics



Threshold
0.5

Point metrics: Confusion Matrix



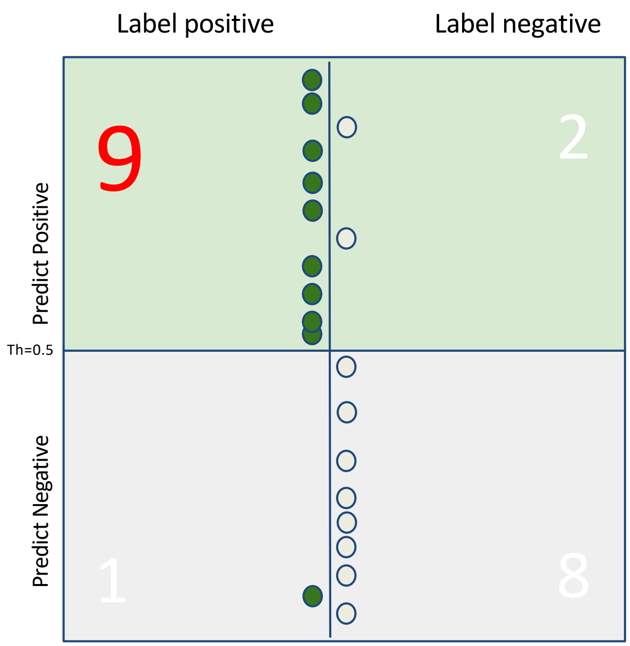
Th
0.5

Properties:

- Total sum is fixed (population).
- Column sums are fixed (class-wise population).
- Quality of model & threshold decide how columns are split into rows.
- We want diagonals to be “heavy”, off diagonals to be “light”.

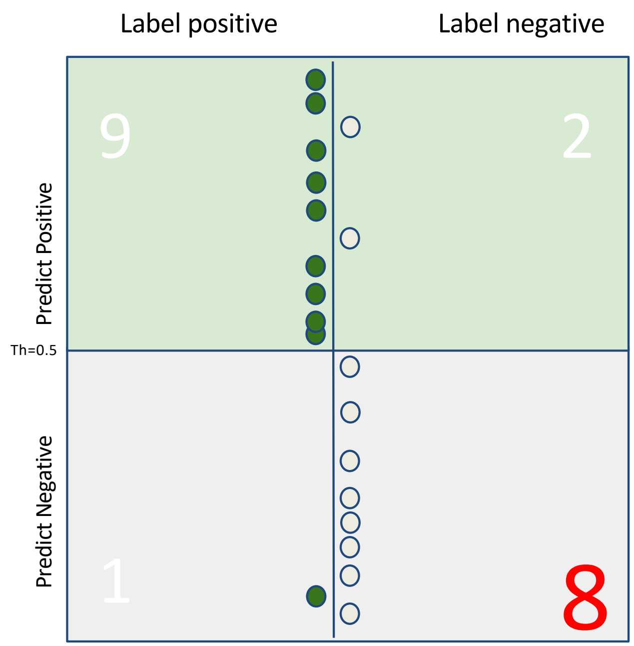
Point metrics: True Positives

True Positives



Th	TP
0.5	9

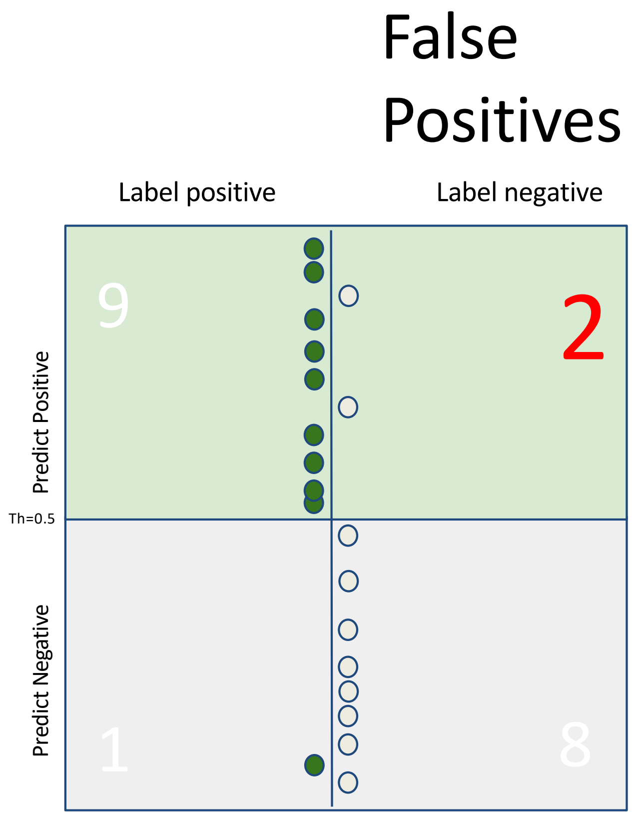
Point metrics: True Negatives



Th	TP	TN
0.5	9	8

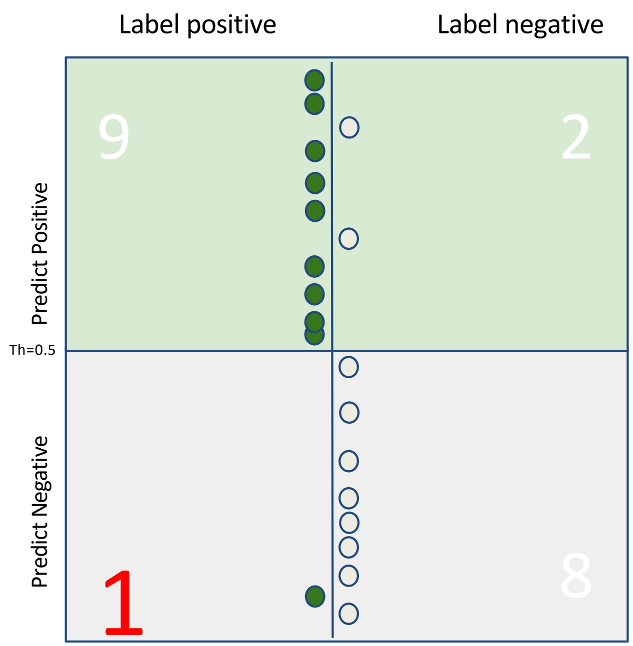
True Negatives

Point metrics: False Positives



Th	TP	TN	FP
0.5	9	8	2

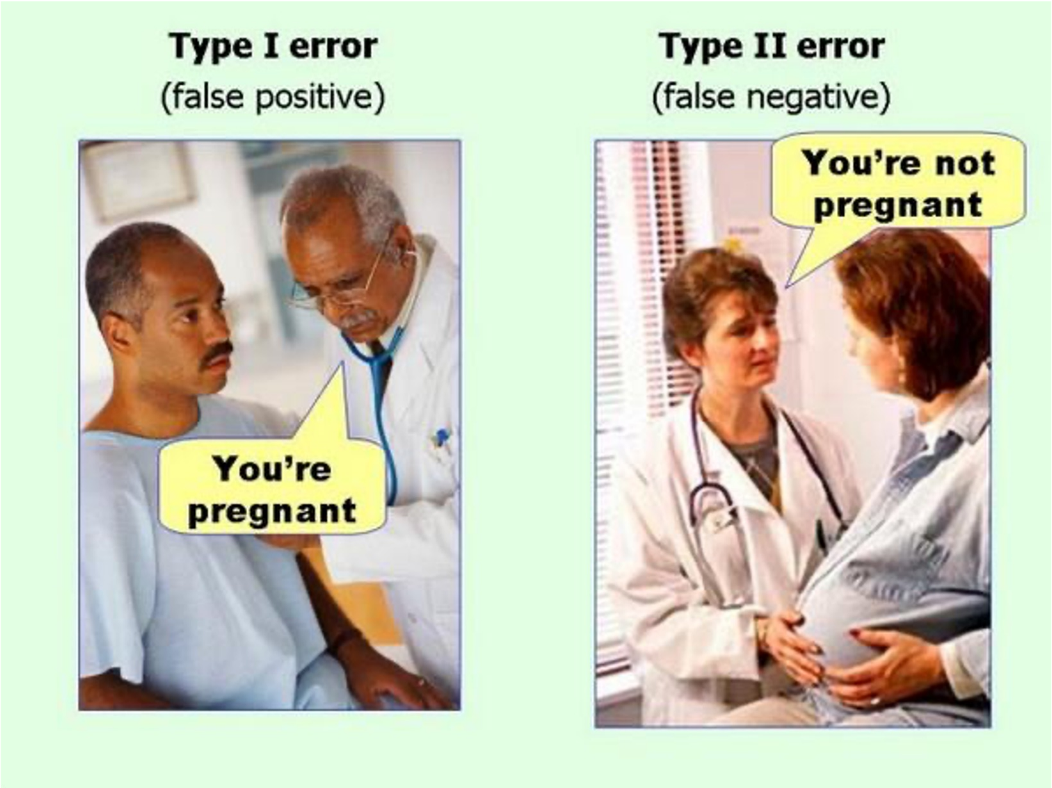
Point metrics: False Negatives



False Negative

Th	TP	TN	FP	FN
0.5	9	8	2	1

FP and FN also called Type-1 and Type-2 errors



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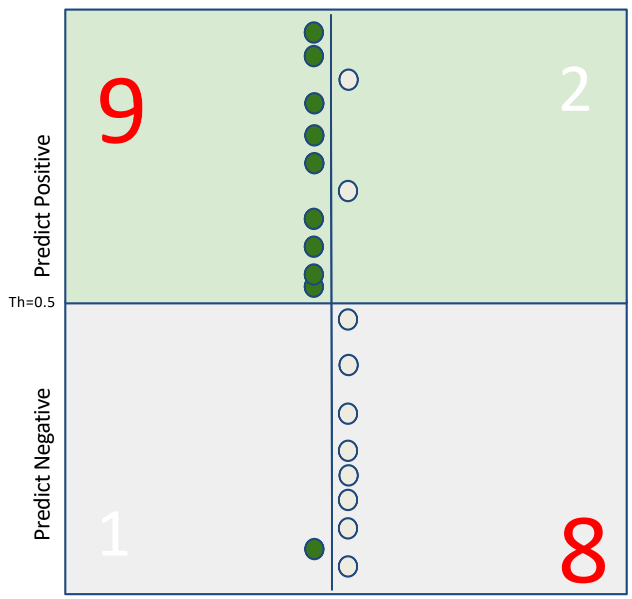
Point metrics: Accuracy

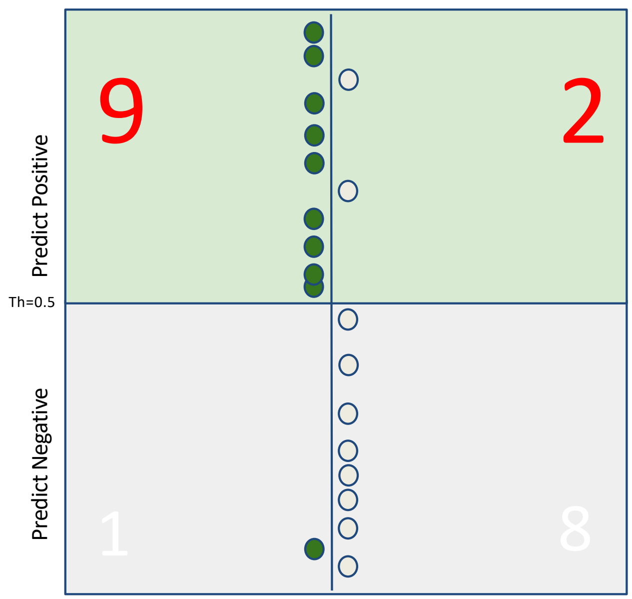


Th	TP	TN	FP	FN	Acc
0.5	9	8	2	1	.85

Accuracy = (TP + TN) / (TP + FP + TN + FN)

Interpretation: overall fraction of correct predictions

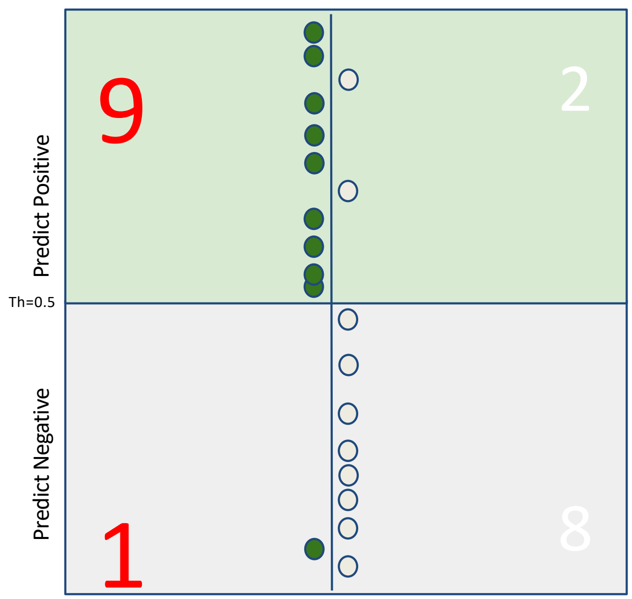
Point metrics: Precision



Th	TP	TN	FP	FN	Acc	Pr
0.5	9	8	2	1	.85	.81

Precision = TP / (TP + FP)
Interpretation: among predicted positives, how many are truly positive?

Point metrics: Positive Recall (Sensitivity)



Th	TP	TN	FP	FN	Acc	Pr	Recall
0.5	9	8	2	1	.85	.81	.9

Recall = TP / (TP + FN)

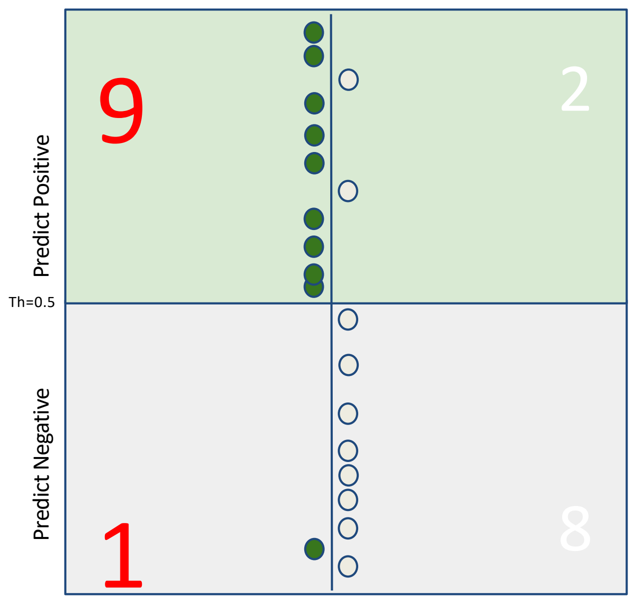
Interpretation: among all true positives, how many did we find?

Trivial 100% recall = pull everybody above the threshold.

Trivial 100% precision = push everybody below the threshold except 1 green on top.

(Hopefully no gray above it!)

Point metrics: Positive Recall (Sensitivity)



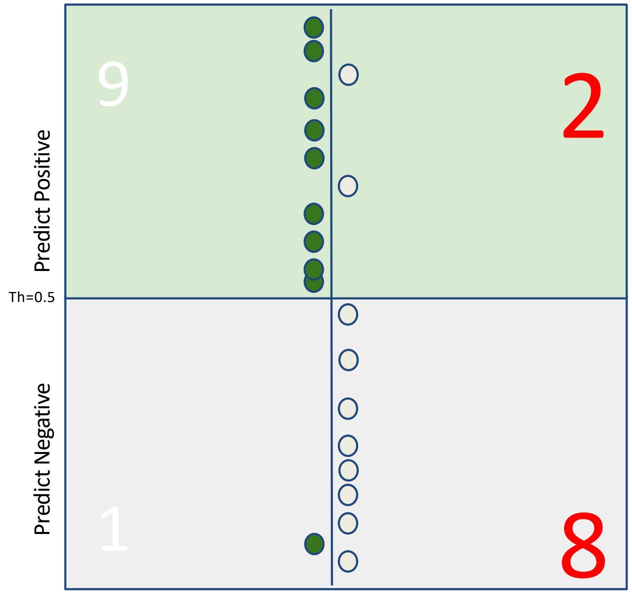
Th	TP	TN	FP	FN	Acc	Pr	Recall
0.5	9	8	2	1	.85	.81	.9

Recall = TP / (TP + FN)
Interpretation: among all true positives, how many did we find?

Striving for good precision with 100% recall =
pulling up the lowest green as high as possible in the ranking.

Striving for good recall with 100% precision =
pushing down the top gray as low as possible in the ranking.

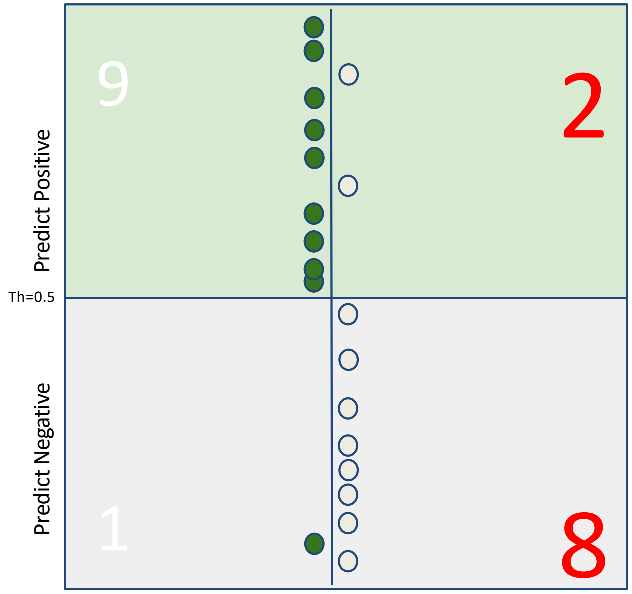
Point metrics: Negative Recall (Specificity)



Th	TP	TN	FP	FN	Acc	Pr	Recall	Spec
0.5	9	8	2	1	.85	.81	.9	0.8

Negative Recall = TN / (TN + FP)
Interpretation: among all true negatives, how many did we find?

Point metrics: False Positive Rate (FPR)



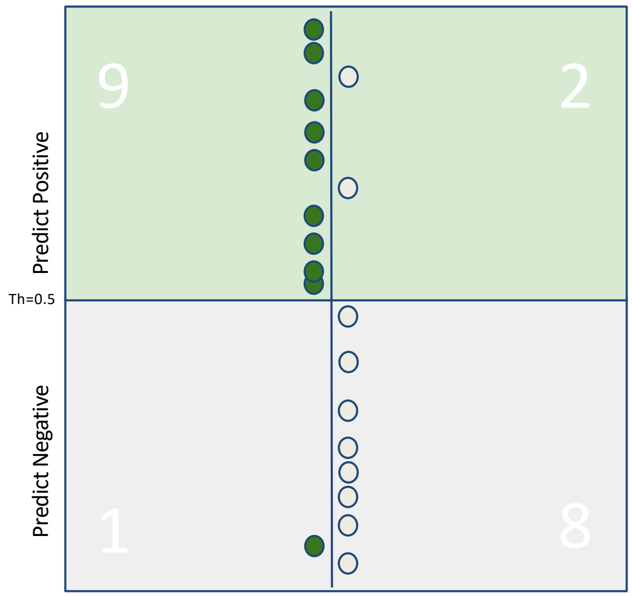
Th	TP	TN	FP	FN	Acc	Pr	Recall	Spec	FPR
0.5	9	8	2	1	.85	.81	.9	0.8	0.2

FPR= FP / (FP + TN)

Interpretation: among all true negatives, how many did we get wrong?

FPR = 1 - Specificity

Point metrics: F1-score



Th	TP	TN	FP	FN	Acc	Pr	Recall	Spec	F1
0.5	9	8	2	1	.85	.81	.9	.8	.857

$$F_1 = \left(\frac{2}{\mathrm{recall}^{-1} + \mathrm{precision}^{-1}}\right) = 2 \cdot \frac{\mathrm{precision} \cdot \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}}.$$

F1-score: Harmonic mean of PR and REC

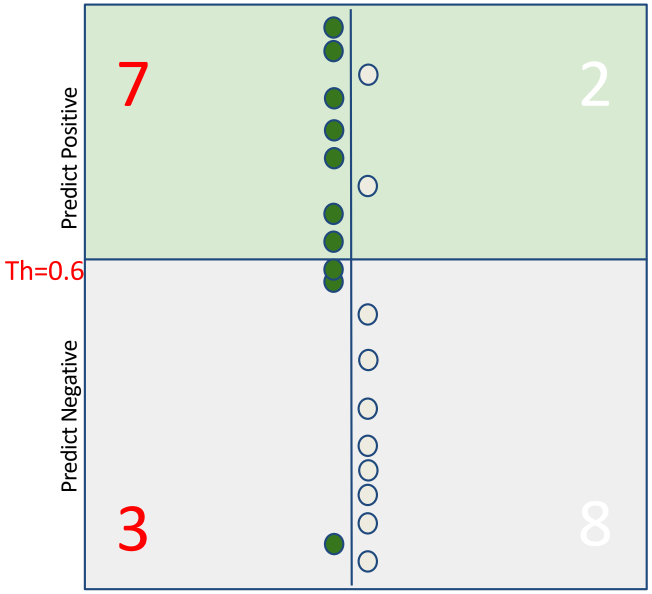
F1 punishes imbalance between precision and recall

Precision	Recall	F1
0	0	0
0.9	0.9	0.9
0.9	0.1	0.18

Accuracy

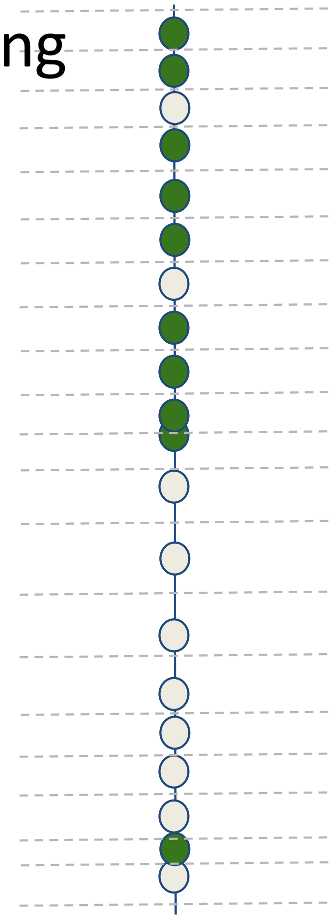
- Accuracy = (TP + TN) / (TP + FP + TN + FN)
  - Interpretation: overall fraction of correct predictions
- Why accuracy can fail (imbalanced data)
  - Example: 10,000 transactions, 100 frauds (1% positives).
  - A dumb model predicting "not fraud" always:
  - Confusion Matrix:
    - TP=0, FP=0, FN=100, TN=9900
  - Accuracy = 99% (looks great)
  - Recall = 0% (catches no fraud)

Point metrics: Changing threshold



Th	TP	TN	FP	FN	Acc	Pr	Recall	Spec	F1
0.6	7	8	2	3	.75	.77	.7	.8	.733

Threshold Scanning



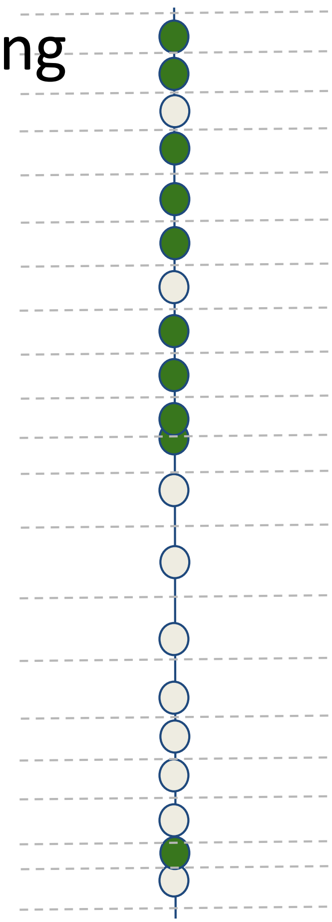
Threshold	TP	TN	FP	FN	Accuracy	Precision	Recall	Specificity	F1
1.00	0	10	0	10	0.50	0	0	1	0
0.95	1	10	0	9	0.55	1	0.1	1	0.182
0.90	2	10	0	8	0.60	1	0.2	1	0.333
0.85	2	9	1	8	0.55	0.667	0.2	0.9	0.308
0.80	3	9	1	7	0.60	0.750	0.3	0.9	0.429
0.75	4	9	1	6	0.65	0.800	0.4	0.9	0.533
0.70	5	9	1	5	0.70	0.833	0.5	0.9	0.625
0.65	5	8	2	5	0.65	0.714	0.5	0.8	0.588
0.60	6	8	2	4	0.70	0.750	0.6	0.8	0.667
0.55	7	8	2	3	0.75	0.778	0.7	0.8	0.737
0.50	8	8	2	2	0.80	0.800	0.8	0.8	0.800
0.45	9	8	2	1	0.85	0.818	0.9	0.8	0.857
0.40	9	7	3	1	0.80	0.750	0.9	0.7	0.818
0.35	9	6	4	1	0.75	0.692	0.9	0.6	0.783
0.30	9	5	5	1	0.70	0.643	0.9	0.5	0.750
0.25	9	4	6	1	0.65	0.600	0.9	0.4	0.720
0.20	9	3	7	1	0.60	0.562	0.9	0.3	0.692
0.15	9	2	8	1	0.55	0.529	0.9	0.2	0.667
0.10	9	1	9	1	0.50	0.500	0.9	0.1	0.643
0.05	10	1	9	0	0.55	0.526	1	0.1	0.690
0.00	10	0	10	0	0.50	0.500	1	0	0.667

ROC (Receiver Operating Characteristic) Curve

- ROC curve plots: TPR (Recall) vs FPR as threshold varies
  - TPR = TP/(TP+FN) (y-axis), eg. sensitivity, recall
  - FPR = FP/(FP+TN) (x-axis), eg. 1-sensitivity
  - ROC is threshold-independent ranking evaluation.

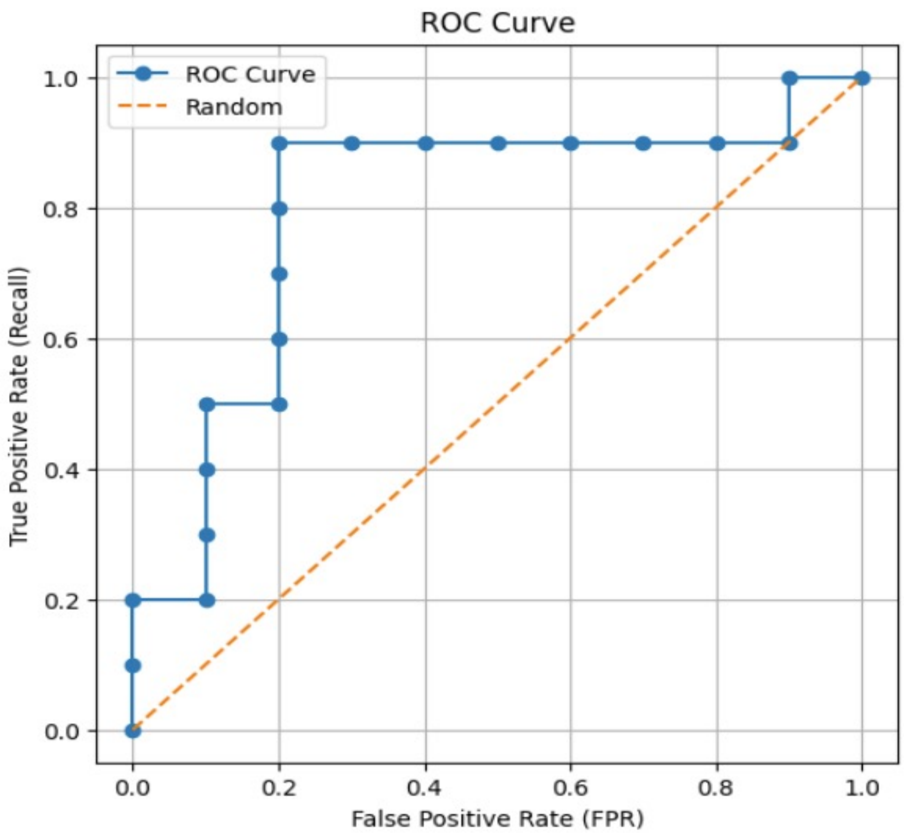
- Some plot ROC as TPR(Recall) vs specificity; that's the same curve with the x-axis reversed since specificity = 1 - FPR
  - TPR(Recall) as y-axis
  - Specificity (TNR) as x-axis

Threshold Scanning



Threshold	TP	TN	FP	FN	Accuracy	Precision	Recall	FPR	Specificity	F1
1.00	0	10	0	10	0.50	0	0	0	1	0
0.95	1	10	0	9	0.55	1	0.1	0	1	0.182
0.90	2	10	0	8	0.60	1	0.2	0	1	0.333
0.85	2	9	1	8	0.55	0.667	0.2	0.1	0.9	0.308
0.80	3	9	1	7	0.60	0.750	0.3	0.1	0.9	0.429
0.75	4	9	1	6	0.65	0.800	0.4	0.1	0.9	0.533
0.70	5	9	1	5	0.70	0.833	0.5	0.1	0.9	0.625
0.65	5	8	2	5	0.65	0.714	0.5	0.2	0.8	0.588
0.60	6	8	2	4	0.70	0.750	0.6	0.2	0.8	0.667
0.55	7	8	2	3	0.75	0.778	0.7	0.2	0.8	0.737
0.50	8	8	2	2	0.80	0.800	0.8	0.2	0.8	0.800
0.45	9	8	2	1	0.85	0.818	0.9	0.2	0.8	0.857
0.40	9	7	3	1	0.80	0.750	0.9	0.3	0.7	0.818
0.35	9	6	4	1	0.75	0.692	0.9	0.4	0.6	0.783
0.30	9	5	5	1	0.70	0.643	0.9	0.5	0.5	0.750
0.25	9	4	6	1	0.65	0.600	0.9	0.6	0.4	0.720
0.20	9	3	7	1	0.60	0.562	0.9	0.7	0.3	0.692
0.15	9	2	8	1	0.55	0.529	0.9	0.8	0.2	0.667
0.10	9	1	9	1	0.50	0.500	0.9	0.9	0.1	0.643
0.05	10	1	9	0	0.55	0.526	1	0.9	0.1	0.690
0.00	10	0	10	0	0.50	0.500	1	1	0	0.667

ROC curve



AUROC = Area Under ROC

= Prob(Random Positive ranked higher than random negative)

ROC measures **how well the model ranks positives above negatives**

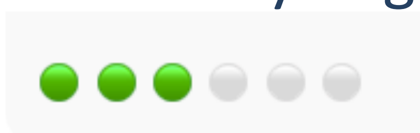
```
from sklearn.metrics import RocCurveDisplay

RocCurveDisplay.from_estimator(model, X_test, y_test)
plt.show()
```

AUROC Curve Intuition (1)

- As we perform rank threshold scanning from t=1 → t=0

  Each time you include:

  - → move **UP** (TPR increases)
  - → move **RIGHT** (FPR increases)

- ROC curve = tracking how many positives you capture vs how many mistakes you make as you go down the ranking
- Perfect model:  UP UP UP → RIGHT RIGHT RIGHT
  - Curve hugs **top-left corner**
  - **Area under the curve AUROC=1**

AUROC Curve Intuition (2)

- The worst model

  - All negatives ranked above positives:
  - Curve goes right first, then straight up → terrible
  - Area under the curve AUROC=0
- The bad (random) model

  - Mixed random ranking
  - Curve zig-zags diagonally whose expected shape is the diagonal
  - Area under the curve AUROC = 0.5

ROC : Agnostic to prevalence!

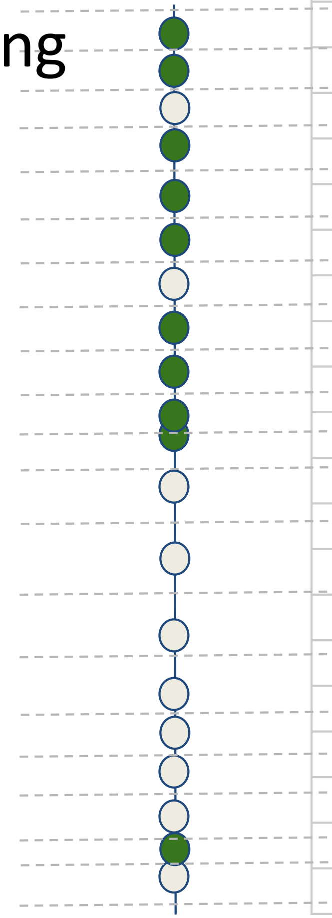
- For imbalanced data i.e., 1% positive / 99% negative → rare event, metrics such as precision fails (Predict all negative → 99% accuracy (useless))

- ROC only uses:
  - TPR = TP / (TP + FN)
  - FPR = FP / (FP + TN)

  Both are **normalized within each class**
- ROC treats:
  - positives independently
  - negatives independently

  So it doesn’t care how many of each exist.

Threshold Scanning



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0.95	1	10	0	9	0.55	1	0.1	0	1	0.182
0.90	2	10	0	8	0.60	1	0.2	0	1	0.333
0.85	2	9	1	8	0.55	0.667	0.2	0.1	0.9	0.308
0.80	3	9	1	7	0.60	0.750	0.3	0.1	0.9	0.429
0.75	4	9	1	6	0.65	0.800	0.4	0.1	0.9	0.533
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0.55	7	8	2	3	0.75	0.778	0.7	0.2	0.8	0.737
0.50	8	8	2	2	0.80	0.800	0.8	0.2	0.8	0.800
0.45	9	8	2	1	0.85	0.818	0.9	0.2	0.8	0.857
0.40	9	7	3	1	0.80	0.750	0.9	0.3	0.7	0.818
0.35	9	6	4	1	0.75	0.692	0.9	0.4	0.6	0.783
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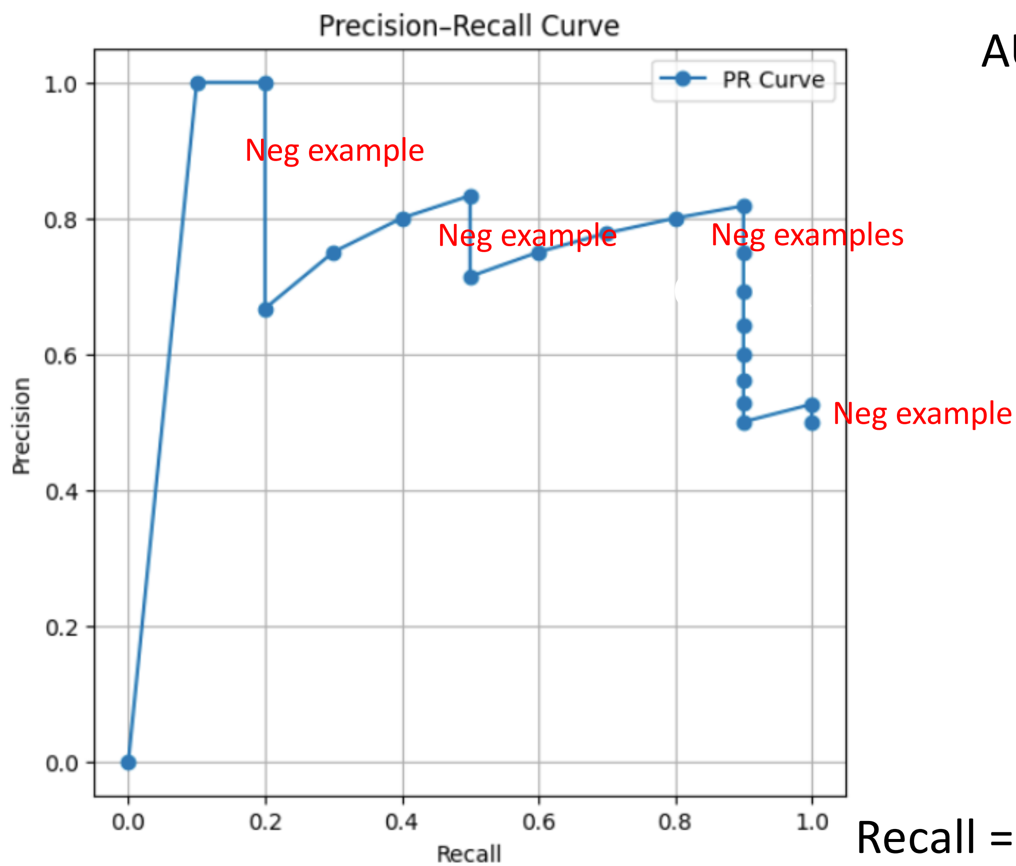
0.25	9	4	6	1	0.65	0.600	0.9	0.6	0.4	0.720
0.20	9	3	7	1	0.60	0.562	0.9	0.7	0.3	0.692
0.15	9	2	8	1	0.55	0.529	0.9	0.8	0.2	0.667
0.10	9	1	9	1	0.50	0.500	0.9	0.9	0.1	0.643
0.05	10	1	9	0	0.55	0.526	1	0.9	0.1	0.690
0.00	10	0	10	0	0.50	0.500	1	1	0	0.667

Precision Recall Curve

Precision
= True Pos /
Predicted Pos



AUPRC = Area Under PRC

= Expected precision for
Random threshold

PRC: How many false positives are you introducing while trying to capture positives?

Recall = Sensitivity = True Pos / Pos

PRC Intuition

- Best model  *All positives are ranked above negatives*
  - Precision stays = **1,** Recall increases from 0 → 1
  - Area under the PR curve AUPRC=1
  - find ALL positives without introducing any false positives
- Worst model (reverse ranking) 
  - precision is 0 first, then **very low initially,** improves slowly when positives appear (1/6, 2/7, … 5/10)
  - AUPRC is small (Depends on **prevalence,** No fixed worst-case value)
- Random model 
  - Roughly a **horizontal line:** AUPRC ≈ prevalence

PRC vs ROC

PR curve behavior is asymmetric:

Add sample	Effect
Positive	Recall ↑, Precision often stable
Negative	Precision ↓, Recall unchanged

ROC curve behavior is more symmetric:

Add sample	Effect
Positive	**TPR ↑**, FPR unchanged
Negative	**FPR ↑**, TPR unchanged

In ROC:

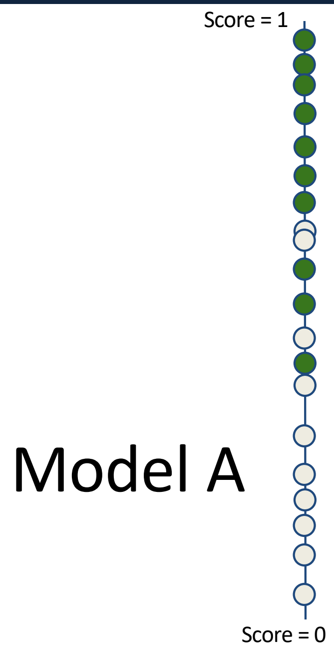
- Adding a negative → small FPR increase (often looks harmless)

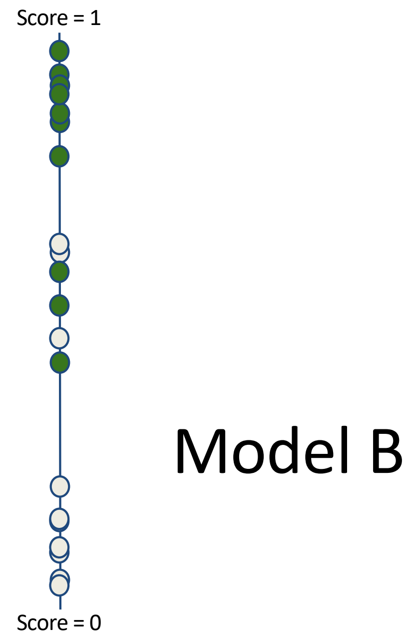
In PRC:

- Adding a negative → **precision drops sharply**

Much more sensitive

Other metrics?





- Two models classifying the same data set with the same rankings :
    - Same ROC curve, PR curve and Accuracy!
    - Is one of them better than the other?

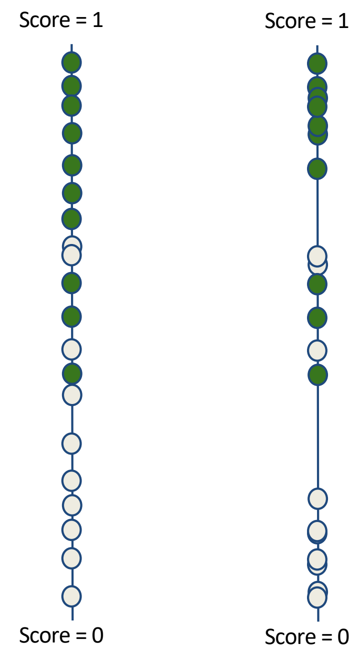
Summary (probability) metrics: Log-Loss vs Brier Score

$$\text{Log Loss} = \frac{1}{N}\sum_{i=1}^{N} -y_i \log \hat{y}_i - (1 - y_i)\log(1 - \hat{y}_i).$$

- Rewards confident correct answers, heavily penalizes confident wrong answers.
  - Predict 0.99 but true = 0 → huge penalty
- **Proper** scoring rule: Minimized at $\hat{y} = y$

$$\text{Brier Score} = \frac{1}{N}\sum_{i=1}^{N} (\hat{y}_i - y_i)^2$$

  - smoother, less harsh than log loss



Two models can have very different Log Loss and Brier scores.

Calibration vs Discriminative Power

Logistic (th=0.5):

Precision: 0.872

Recall: 0.851

F1: 0.862

Brier: 0.099 ← better

similar discrimination

SVC (th=0.5):

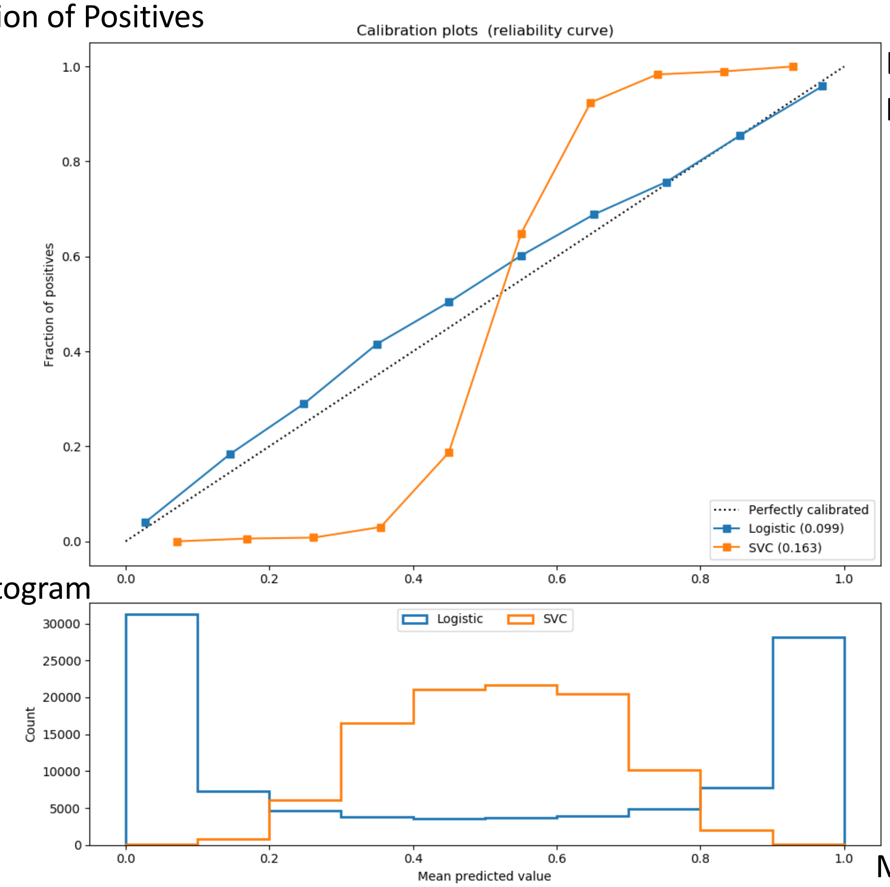
Precision: 0.872

Recall: 0.852

F1: 0.862

Brier: 0.163

Fraction of Positives



Histogram

Diagonal means perfect calibration.

Calibration: Are the predicted probabilities numerically correct?

Mean predicted value

Multi-class

- Confusion matrix will be N * N (still want heavy diagonals, light off-diagonals)
- Most metrics (except accuracy) generally analyzed as multiple One-vs-rest
- Multiclass variants of AUROC and AUPRC (micro vs macro averaging)
- Class imbalance is common (both in absolute and relative sense)
  - Cost sensitive learning techniques (also helps in binary imbalance)
    - Assign weights for each block in the confusion matrix.
    - Incorporate weights into the loss function.

Multiclass confusion matrix

Example: 3-class medical triage (Low, Medium, High risk). Classes:

- 0 = Low
- 1 = Medium
- 2 = High

Asymmetric mistake costs (missing "High" is very bad): $W = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 2 \\ 10 & 5 & 0 \end{bmatrix}$

Interpretation:

- True High (2) predicted as Low (0): cost = 10 (very bad)
- True High predicted as Medium: cost = 5
- True Low predicted as High: cost = 2 (annoying but less catastrophic)

Suppose the confusion matrix (counts) is:

$$N = \begin{bmatrix} 50 & 8 & 2 \\ 6 & 30 & 4 \\ 3 & 5 & 12 \end{bmatrix}$$

Rows = actual, columns = predicted.

Multiclass confusion matrix

- **Weighted total cost (weighted error)**

$$\text{TotalCost} = \sum i = 0 \overset{j=0}{\underset{\sum}{2}} 2\, N_{i,j}\, W_{i,j}$$

Compute it (ignore diagonal since cost=0):

- Row Low (0): $8 \cdot 1 + 2 \cdot 2 = 8 + 4 = 12$
- Row Med (1): $6 \cdot 1 + 4 \cdot 2 = 6 + 8 = 14$
- Row High (2): $3 \cdot 10 + 5 \cdot 5 = 30 + 25 = 55$

TotalCost = 12 + 14 + 55 = 81

So even if overall accuracy looks “fine”, the model is getting **expensive mistakes** on class 2.

- **Cost per sample** :

$$\text{AvgCost} = \frac{81}{\sum_{i,j} N_{i,j}}$$

Total samples $= 50 + 8 + 2 + 6 + 30 + 4 + 3 + 5 + 12 = 120.$
So **AvgCost = 81/120 = 0.675**

Metrics in Multiclass

- Convert to one-vs-rest evaluation:
  - For 3 class problem: A, B, C
  - Form 3 data:
    - For class A: A vs (B+C), i.e., A→ positive(1), B,C→ negative(0)
    - For class B: B vs (A+C), i.e., B→positive(1), A,C→negative (0)
    - For class C: C vs (A+B), i.e., C→ positive(1), B, C→ negative(0)
- For precision:
  - Macro average
  - Micro average
  - Weighted average

$$\text{Precision}_A = \frac{TP_A}{TP_A + FP_A}$$

$$\text{Macro Precision} = \frac{P_A + P_B + P_C}{3}$$

Metrics in Multiclass

Macro metric

– Compute $\text{Precision}_{A,}$ Precision_{B} and Precision_{C} for class A, B and C independently

$$\text{Precision}_A = \frac{TP_A}{TP_A + FP_A}$$

– Macro Average: $\text{Macro Precision} = \frac{P_A + P_B + P_C}{3}$

Example: true class labels: A, C, A, A, B, A

predicted class labels: A, B, C, A, A, A

Class A

- TP_A = 3 (1,4,6)
- FP_A = 1 (5)
- FN_A = 1 (3)

$$P_A = \frac{3}{3+1} = 0.75, \quad R_A = \frac{3}{3+1} = 0.75$$

Class B

- TP_B = 0
- FP_B = 1 (2)
- FN_B = 1 (5)

$$P_B = 0, \quad R_B = 0$$

Class C

- TP_C = 0
- FP_C = 1 (3)
- FN_C = 1 (2)

$$P_C = 0, \quad R_C = 0$$

Macro Precision=(0.75+0+0)/3=0.25, Macro Recall = 0.75+0+0)/3=0.25

Metrics in Multiclass

Micro Average

- Aggregate all TP/FP/FN first, then compute metric globally

Global counts

- TP = 3 (correct predictions)
- FP = 3 (wrong predictions)
- FN = 3 (missed truths)

$$\text{Micro Precision} = \frac{3}{3+3} = 0.5$$

$$\text{Micro Recall} = \frac{3}{3+3} = 0.5$$

Metrics in Multiclass

Weighted Average

- Weight by class size: $P_{weighted} = \frac{\sum_k support(k)\, P_k}{\sum_k support(k)}$, $R_{weighted} = \frac{\sum_k support(k)\, R_k}{\sum_k support(k)}$

The weights are class supports (from true class labels)

For the example, w_A=4, w_B=1, w_C=1

Weighted Precision

$$P_w = \frac{4 \cdot 0.75 + 1 \cdot 0 + 1 \cdot 0}{6} = \frac{3}{6} = 0.5$$

Weighted Recall

$$R_w = \frac{4 \cdot 0.75 + 1 \cdot 0 + 1 \cdot 0}{6} = \frac{3}{6} = 0.5$$

ROC and PRC in Multiclass

- Approach One: Turn Multiclass → Multiple Binary Problems
- Example: given 3-class classification (A, B, C),
  – for class A:
    - treat A as positive, B and C as negative,
    - Perform threshold scan from p(A) = 1.0 to 0.0, record TP, FP, FN, TN for each threshold value
    - Plot FPR vs FPR ➔ gives ROC curve for class A
  – Repeat the steps for class B and class C separately to obtain ROC curves for the two classes
  – PRC curves can be obtained in the same approach

ROC and PRC in Multiclass

- Approach Two: Macro Averaging
  - Average the ROC (or PR) curves of individual classes (e.g. by interpolation)
- Approach Three: Micro Averaging
  - Directly perform multiclass classification prediction, i.e., do not convert into binary classification problems
  - Compute the global TP, FP, FN, TN values
  - Plot one ROC or PR curve based on the global values

Unsupervised Learning

- Log P(x) is a measure of fit in Probabilistic models (GMM, Factor Analysis)
  - High log P(x) on training set, but low log P(x) on test set is a measure of overfitting
  - Raw value of log P(x) hard to interpret in isolation

- K-means is trickier (because of fixed covariance assumption)

Choosing Metrics

- Care about ranking or decisions?
  - Rank examples (who is more likely?) : AUROC, AUPRC
  - Make decision (classify): Accuracy, Precision, Recall, F1
  - Output good probabilities: Log Loss, Brier
- Balanced vs imbalanced data:
  - Balanced: Accuracy, AUROC
  - Imbalanced: PR curve/ AUPRC, Precision, Recall, F1
- Need to compute probabilities?
  - Yes: log loss, Brier Score, Calibration curves
  - No: Precision, Recall, F1, Accuracy

Choosing Metrics

- High precision is hard constraint, do best recall (search engine results, grammar correction): Intolerant to FP
  - Metric: Recall at Precision = XX %
- High recall is hard constraint, do best precision (medical diagnosis, fraud detection): Intolerant to FN
  - Metric: Precision at Recall = 100 %
- Both matters, choose F1 score
- Capacity constrained (by K)
  - Metric: Precision in top-K.

Choosing Metrics – Practical Cheat Sheet

- Binary classification:
  - Balanced: ROC AUC + Accuracy
  - Imbalanced: AUPRC + F1
- Risk/probability tasks:
  - Log loss, Brier, Calibration Curve
- Multiclass:
  - Accuracy, Macro F1, Micro F1,

Metrics Use in Practice

Metric	Usage
Accuracy	baseline
AUROC	very common
AUPRC	rare events
F1	classification tasks
Log Loss	training + evaluation
Brier	calibration analysis